

FERGUS FALLS, MN, USA

WMO: 726572

Lat: 46.283N

Lon: 96.150W

Elev: 1183

StdP: 14.08

Time Zone: -6.00 (NAC)

Period: 96-19

WBAN: 94966

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.5	-15.1	-26.3	1.3	-19.3	-20.3	1.9	-13.7	34.7	7.6	31.2	8.7	6.4	320	0.622

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.7	88.4	72.1	84.5	70.2	82.1	68.9	75.7	84.1	73.8	81.5	71.8	79.6	13.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.9	127.5	81.1	71.8	122.8	79.9	69.8	114.5	77.4	40.2	84.3	38.1	81.1	36.2	78.7	84.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.6	24.6	21.3	DB	-26.4	93.9	5.4	3.9	-30.3	96.7	-33.5	99.0	-36.5	101.2	-40.4	104.1	(4)
(5)			WB	-26.7	79.7	5.3	2.1	-30.5	81.2	-33.7	82.5	-36.6	83.7	-40.5	85.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	43.1	11.0	12.8	28.7	43.6	57.1	67.3	71.8	68.9	61.2	46.5	30.7	15.4		
	DBStd	24.22	14.85	13.49	13.58	10.86	9.31	6.35	5.49	5.45	8.28	9.49	12.20	12.96		
	HDD50	5049	1210	1041	669	243	40	0	0	0	8	182	584	1072		
	HDD65	8575	1675	1461	1126	642	275	49	10	22	170	579	1029	1537		
	CDD50	2524	0	0	8	52	260	518	677	587	344	73	5	0		
	CDD65	576	0	0	0	1	30	117	223	144	56	5	0	0		
	CDH74	4573	0	0	1	18	331	928	1813	1012	422	48	0	0		
	CDH80	1240	0	0	0	3	102	239	538	245	103	10	0	0		
	WSAvg	9.5	10.0	9.6	10.2	10.8	10.6	9.1	7.7	7.6	9.0	10.2	10.2	9.5		
	PrecAvg	23.1	0.7	0.6	1.1	2.1	2.7	3.8	3.3	3.1	2.4	1.7	0.9	0.7		
(15) Precipitation	PrecMax	35.6	3.1	1.4	2.2	4.9	7.0	9.5	7.3	6.9	6.6	6.3	4.4	2.4		
	PrecMin	15.1	0.1	0.0	0.2	0.0	0.5	0.7	0.4	0.2	0.3	0.1	0.1	0.0		
	PrecStd	5.0	0.7	0.4	0.6	1.3	1.8	1.9	1.7	1.5	1.7	1.5	0.9	0.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.5	47.9	66.1	75.3	89.5	90.8	92.9	90.2	88.1	80.8	63.6	45.8		
		MCWB	36.7	44.4	54.6	57.2	66.9	73.1	76.3	71.5	69.8	63.6	53.0	41.0		
	2%	DB	36.5	39.1	58.9	70.2	82.0	86.0	89.5	85.6	82.4	72.3	57.1	37.4		
		MCWB	33.8	36.1	49.8	53.5	64.5	69.8	74.5	72.5	68.3	58.1	48.2	34.2		
	5%	DB	34.0	35.8	52.4	65.2	77.0	81.9	84.5	82.0	79.4	65.9	52.1	35.7		
		MCWB	32.3	33.8	46.1	51.7	61.7	67.3	71.7	69.9	66.6	55.7	45.7	33.7		
	10%	DB	30.1	31.5	45.5	61.2	72.4	79.4	82.1	80.6	74.8	61.1	46.0	32.2		
		MCWB	29.4	30.5	40.9	49.1	59.5	65.6	70.2	68.7	65.0	52.1	41.3	31.1		
	0.4%	WB	37.2	44.1	58.8	59.7	71.3	77.4	80.0	76.7	74.8	65.1	55.2	41.5		
		MCDB	41.5	46.6	63.5	71.3	80.3	84.6	89.2	84.2	81.3	75.5	62.4	46.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	34.1	36.9	51.8	56.0	67.4	73.4	76.6	74.4	72.0	60.2	49.9	35.8		
		MCDB	36.1	39.2	57.1	65.8	77.2	81.8	85.9	81.9	79.1	69.7	56.1	37.7		
	5%	WB	32.0	33.4	45.8	53.1	64.4	70.5	74.5	72.6	68.8	56.6	45.7	33.4		
		MCDB	33.6	35.2	50.9	62.2	73.1	78.9	82.9	79.5	76.7	64.4	50.7	35.1		
	10%	WB	29.4	29.9	40.7	50.2	61.8	67.9	72.2	70.6	66.2	53.0	41.7	30.6		
		MCDB	29.9	31.2	46.1	59.7	69.7	76.3	80.5	77.2	73.9	59.7	46.6	31.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	17.5	17.8	17.0	19.6	20.8	19.0	19.7	19.2	19.6	17.4	15.7	15.3		
		MCDBR	18.8	19.5	23.8	28.1	27.3	22.6	22.9	21.9	24.0	24.5	22.7	17.3		
		MCWBR	15.7	16.1	15.7	15.8	14.3	11.4	11.8	10.8	12.6	14.1	14.8	14.2		
	5% WB	MCDBR	18.0	19.5	23.0	24.9	23.1	20.0	20.2	19.4	20.3	21.8	20.8	16.1		
		MCWBR	15.2	16.5	15.8	15.5	13.7	12.4	12.0	11.1	12.4	14.1	14.2	13.8		
(40) Clear-Sky Solar Irradiance	taub		0.282	0.291	0.303	0.350	0.370	0.380	0.381	0.382	0.351	0.313	0.284	0.264		
	taud		2.317	2.316	2.429	2.360	2.372	2.384	2.376	2.372	2.447	2.534	2.515	2.417		
	Ebn at Noon		258	280	295	287	284	281	279	274	274	272	261	256		
	Edh at Noon		24	30	31	36	37	37	37	36	30	24	20	19		
(44) All-Sky Solar Radiation	RadAvg		531	887	1218	1545	1734	1927	2034	1706	1290	806	507	377		
	RadStd		58	95	118	122	122	122	82	150	88	115	55	43		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (37 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page